

Shujie XU

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Shanghai, China

EDUCATION

- **Northwestern University & Chinese University of Hong Kong** Aug. 2024 - Dec. 2025
Dual M.S. in Social & Economic Policy and Applied Economics Evanston, USA & Hong Kong SAR
 - **GPA:** Northwestern 3.89/4.00 | CUHK 3.83/4.00
 - **Selected Coursework:** Macroeconomic Analysis & Applications (A, Excellent Performance Award), Microeconomic Analysis & Applications (A), Econometric Analysis & Applications (A), Machine Learning for Economists (A), Causal Methods for Policy Evaluation, Mathematics for Economic Analysis
- **Cambridge University** Jul. 2023 - Aug. 2023
Summer Academic Program - Quantitative Finance & Investment Track Cambridge, United Kingdom
 - **Grade:** A+ (93.6/100, Top 5%); **Honors:** Best Project Award | Supervisor: Prof. Andrei Kirilenko
- **Southwestern University of Finance and Economics** Sep. 2020 - Jul. 2024
Bachelor of Economics in Finance (Major), Economics (Minor) Chengdu, China
 - **GPA:** 3.60/4.00; **Average Score:** 85.00/100
 - **Selected Coursework:** Advanced Microeconomics, Intermediate Macroeconomics, Econometrics, Advanced Algebra, Programming and Python
 - **Thesis:** *Political Attention, Partisan Alignment, and Corporate Performance* | Supervised by Prof. Yue TANG

RESEARCH INTERESTS

- My research studies how information frictions and institutional contexts shape firm/individual behavior and outcomes. I use computational methods including ML/NLP and network analysis to extract insights from unstructured data, to examine strategic decision-making in organizational and market settings.
- **Primary Fields:** Applied Microeconomics, Corporate Finance, Urban Economics
- **Research Areas:** Information economics; corporate political engagement; housing markets
- **Methods:** Natural language processing, causal inference, computational economics

RESEARCH EXPERIENCES

- **Predoctoral Research Fellow** Jul. 2026 - Present
Northwestern University, Kellogg School of Management Evanston, USA / Full-time
- **Research Assistant to Prof. Fan Wu** Mar. 2026 - Jun. 2026
The Chinese University of Hong Kong Hong Kong SAR / Full-time
 - Project:** Regulation in analyst reports
 - **Designed measurement strategy for regulatory discussion in analyst reports.** Translated conceptual definitions into GPT prompts and validation rules to identify specific regulation mentions from a large analyst-report corpus (3.7M paragraphs). Distinguished specific rules, broad policy references, and regulatory authorities to keep the measure aligned with the paper's construct.
 - **Built interpretable measures of analysts' regulatory interpretation.** Classified roughly 510K regulation mentions by category, sentiment, uncertainty, discussion type, and valuation channel; used these outputs to study when regulation is framed as compliance cost, legal risk, demand/pricing shock, or strategic condition.
 - **Developed initial empirical narrative for the regulation report draft.** Compared category composition, time-series patterns, and cross-category heterogeneity; linked report-level regulation language to firm and analyst characteristics to guide the initial draft and follow-up tests.
 - Project:** Corporate strategy measurement in analyst reports
 - **Explored dictionary-based measurement of corporate strategy concepts.** Mapped strategy constructs into six dimensions, then used seed dictionaries and Word2Vec expansion to assess whether analyst-report language can support firm-time measures of corporate strategy.
 - Project:** Supply-chain transparency and patent litigation risk
 - **Extended robustness logic for the supply-chain transparency project.** Evaluated alternative litigation-risk measures, disclosure outcomes, and partner-level exposure to clarify which mechanisms are supported by the current evidence.

• Research Assistant to Prof. Franklin Qian

University of North Carolina, Kenan-Flagler Business School

Mar. 2025 - Jun. 2026

Remote / Part-time

Project: "The Numbers Game: Effects of Listing Pricing Formats in Housing Bargaining"

by Haaris Mateen, Franklin Qian, Ye Zhang, and Tianxiang Zheng

- **Designed text-based measure of seller impatience to empirically test paper's theoretical sorting mechanism.** Proposed and implemented measurement strategy using Word2Vec embeddings on 80M+ property listing remarks from a large proprietary multi-market housing transactions dataset to identify sellers likely to select round vs. precise pricing. Developed domain-specific seed words and preprocessing pipeline; validated measure against human-coded benchmark achieving 96% precision at 62% recall. Measure enables direct test of predicted seller sorting patterns.
- **Extended probabilistic matching methods to merge federally mandated mortgage reporting data with proprietary property transaction records.** Developed procedures handling Census tract boundary changes across 2000, 2010, 2020 vintages; achieved up to 65.6% match rate on 100M+ observations while maintaining covariate balance. Documented methodology into comprehensive replication materials now adopted by other research teams at UNC. Linked dataset enables heterogeneity analysis by borrower demographics and transaction characteristics.
- **Designed and executed heterogeneity analysis examining treatment effects across demographic groups and price ranges.** Specified regression framework testing differential price format effects by borrower race, income quartiles, and property value segments. Implemented estimations in Stata with computationally intensive specifications in Julia; conducted robustness checks varying control sets, sample restrictions, and clustering levels.

Revision Project: "How Housing Search and Bargaining Unfold: Evidence from Offer-Level Data"

Stage: Revise-and-resubmit

- **Contributed to the revise-and-resubmit stage of an offer-level housing bargaining manuscript.** Developed empirical checks addressing referee concerns on listing-price revisions, bidder competition, and bargaining dynamics; clarified sample definitions and robustness interpretation for manuscript revisions.

• Research Assistant to Prof. Letian Zhang

Harvard Business School (now Northwestern University, Kellogg School of Management)

Aug. 2024 - Mar. 2025

Remote / Part-time

Project: Corporate behavior and human capital investment under varying political contexts

Building foundational datasets to examine how political institutions and leader characteristics shape firm strategy

Research Contributions:

- **Constructed longitudinal database of 600+ political leaders across 30+ countries with systematic coding of leader characteristics.** Following established classification framework, coded ideological orientation (left/right positioning), governance style (populist/technocratic), and institutional constraints (executive power, term limits, coalition structure). Documented leadership transitions and designed database schema linking leader profiles with temporal political context variables, supporting multiple projects examining political-economy mechanisms.
- **Built text corpus of 9,000+ political speeches spanning diverse contexts and formats for planned text analysis.** Collected speeches from government archives, legislative records, and media outlets including policy addresses, press conferences, and international forums. Developed automated transcription pipeline for audio/video materials with quality controls for variable audio quality and multilingual content. Structured metadata organized in panel format to enable future supervised learning and topic modeling applications.

• Research Assistant to Prof. Yue Tang

Southwestern University of Finance and Economics

Feb. 2023 - Jan. 2024

Chengdu, China / Part-time

Project: How does firm position in supply chain networks affect exposure to external shocks?

Working Paper: "Enterprise Supply Chain Risk and Network Status: Evidence from Annual Report Text Mining"

Role: Co-authored full initial draft; independently designed and executed data construction, text analysis, and econometric estimation

Research Contributions:

- **Designed text mining pipeline extracting supply chain risk measures from 10,000+ Chinese corporate annual reports (2012-2021).** Developed Chinese-language keyword dictionaries and dependency parsing rules to identify risk-related disclosures; constructed firm-year level risk intensity measures. Validated measure against external shocks including 2020 U.S.-China trade disruptions. Text processing implemented using jieba and gensim; pipeline design enables extension to other disclosure contexts.
- **Constructed supply chain network from disclosed supplier-customer relationships and computed multiple centrality measures.** Built directed network graph from transaction data covering all A-share listed firms; calculated PageRank, eigenvector centrality, and closeness centrality. Network measures test hypothesis that central firms face lower risk exposure through diversification and enhanced bargaining power, controlling for firm size and industry composition.
- **Conducted full empirical analysis using fixed effects panel regressions with firm and industry-year fixed effects.** Implemented rolling window regressions examining temporal variation in centrality-risk relationship. Findings indicate central firms exhibit 16% lower risk exposure with magnified effects during aggregate uncertainty periods. Presented results at departmental research seminar; incorporated feedback into manuscript revisions.

RESEARCH PROJECTS

• **Political Attention, Partisan Alignment, and Corporate Performance**

Undergraduate Thesis | Advisor: Prof. Yue Tang

Research Question: How do corporate political characteristics measured through earnings call discourse affect firm performance?

Research Contributions:

- **Developed "policy-taker" framework where firms respond to political climate through disclosure rather than direct connections.** Theory predicts Political Attention enhances performance via lobbying mediation and reputation mechanisms, moderated by Partisan Alignment. Addresses limitations of CEO-level political proxies with agency problems and selection bias.
- **Constructed text-based measures using semi-supervised ML on 220M+ earnings calls and 3K+ presidential documents.** Built Political Terms dictionary; extracted Presidential Political Terms using TF-IDF and categorized into policy themes; applied Word2Vec to identify synonyms addressing political-corporate vocabulary mismatch.
- **Validated theory through fixed effects regressions: 1% increase in Political Attention yields 2.0pp ROA improvement.** Findings show 15.8% of effect mediated via lobbying expenditure; Partisan Alignment adds 2.8pp for aligned firms. Policy-relevant industries (healthcare, energy, finance) show 0.4pp stronger sensitivity. Results robust to alternative performance measures (ROE: 4.2pp; Sales: 3.3pp).

Nov. 2023 - May 2024

Chengdu

PROFESSIONAL EXPERIENCE

• **Risk Management Analyst Intern**

Financial risk assessment and portfolio analysis

Industrial Research Assistant

Industry analysis and company research

Field Interviewer & Research Assistant

Household survey data collection and processing

Oct. 2023 - Jan. 2024

Sinolink Securities

Jan. 2023 - May 2023

China Securities

May 2022 - Aug. 2022

Survey and Research Center for China Household Finance

SKILLS

- **Programming Languages:** Python, Stata, MATLAB, Julia, R, SQL, Shell/Bash
- **Python Libraries:** pandas, NumPy, scikit-learn, gensim, NLTK
- **Tools & Environments:** Git/GitHub, LaTeX, Quarto, VS Code, High-Performance Computing, Linux/Unix
- **Languages:** English (fluent), Mandarin (native), Spanish (intermediate)